INCH-POUND

MIL-C-70845(AR)
25 July 1989

MILITARY SPECIFICATION

CARTRIDGE, 9mm, PRACTICE, TRACER, M939

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, firing and quality assurance requirements for the testing, inspection and packaging for the 9mm, Tracer, Training Cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification for
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets, Preparation and Submission of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-636 - Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packing, Packaging and Marking of Small Arms Ammunition
MIL-STD-650 - Explosives: Sampling, Inspection and Testing
MIL-STD-810 - Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-1168 - Lot Numbering of Ammunition
DOD-STD-1468 - Small Caliber Ammunition Test Procedures 9mm Cartridges
MIL-STD-1751 - Safety and Performance Test for Qualification of Explosive

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

7643674 - Classification of Cartridge Case Defects for Small Arms Ammunition
D12913886 - Cartridge, 9mm, Practice, Tracer, M939
LI12913886 - Inspection Equipment Lists for Cartridge, 9mm, Practice, Tracer, M939
F12913959 - Packaging and Marking, Cartridge, 9mm, Practice, Tracer, M939, Box, Wirebound
F12913960 - Packaging and Marking, Cartridge, 9mm, Practice, Tracer, M939, Box, M2A2
F12913962 - Packaging and Marking, Cartridge, 9mm, Practice, Tracer, M939, Carton

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PUBLICATIONS

DEPARTMENT OF DEFENSE

TB 700-2	- Department of Defense Explosive Hazard Classification Procedures
TOP 1-2-601	- Laboratory Vibration Schedules, TECOM Test Operating Procedures
TOP 4-2-602	- Rough Handling, TECOM Test Operating Procedures
TOP 2-2-614	- Toxic Fumes Test, TECOM Test Operating Procedures

DEPARTMENT OF INTERIOR

Bureau of Mines
Report S624 - Laboratory Equipment and Test Procedures for Evaluating Explosibility

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, (except for associated detail specifications, specifications sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3.0 REQUIREMENTS

3.1 General. The cartridges shall comply with the requirements shown on drawing 12913886, MIL-A-2550, all other applicable specifications and the requirements stated in this section. All of the cartridge components shall be compatible with each other. The primer mix and pyrotechnic mix shall be non-corrosive and the propellant shall be smokeless. The cartridge shall be compatible with the sub-caliber barrel and bolt assembly of the M287 Sub-Caliber Trainer for the AT4 Weapons System.

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3.2 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (See 4.3).

3.3 Performance characteristics.

3.3.1 Primer sensitivity. The energy imparted by a 55 ± 0.5 gram steel ball falling 12.0 inches shall initiate all primers. The energy imparted by a 55 ± 0.5 gram steel ball falling three inches shall not cause initiation of the primer.

3.3.2 Residual stress. The cartridge shall not split when subjected to the accelerated stress test for the applicable materials.

3.3.3 Bullet extraction. The force required to seperate the bullet from the cartridge case shall be not less than 60 pounds.

3.3.4 Waterproof (bubble test). The cartridge shall release not more than one bubble of air when tested at a pressure of two (2) pounds per square inch (psi) below atmospheric pressure for thirty (30) seconds.

3.3.5 Chamber pressure. The corrected average peak chamber pressure of the cartridge conditioned at 21°C and measured with a piezoelectric gauge at the mid case position shall not exceed 159 Megapascals (MPa) (23,000 psi). No individual peak pressure shall exceed 190 MPa (27,550 psi). When conditioned and fired at the following temperatures no individual peak chamber pressure shall exceed 190 MPa (27,550 psi).

-18°C ± 2°C
60°C ± 2°C

3.3.6 Dispersion. The average linear standard deviation, both horizontal and vertical, for all ten round targets fired at a range of 100 meters shall not exceed 4.3 inches.

3.3.7 Function and casualty. The cartridge shall safely function in the M287 sub-caliber training weapon without casualty in the temperature range of -18°C to 60°C.

3.3.8 Trajectory. The trajectory of the projectile shall match, within ± 1 mil, the trajectory of the AT4 tactical round, M136, at all ranges out to 300 meters. The firing table for the AT4 round is given in Table I. Trajectory assessment shall be conducted for first article only. The velocity requirement established as part of first article testing will control trajectory match for production ammunition.

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TABLE I. AT4 firing table for standard conditions

Range (Meters)	Time of flight (seconds)	Maximum ordinate (Meters)	Range to maximum ordinate (Meters)
0	0.0	0.0	0
50	0.2	0.0	25
100	0.4	0.2	51
150	0.5	0.4	77
200	0.7	0.7	103
250	1.0	1.1	130
300	1.2	1.7	157
350	1.4	2.5	185
400	1.7	3.4	212
450	1.9	4.5	241
500	2.2	5.8	269

3.3.9 Velocity. The contractor will define a design velocity and tolerance (at a point 16 meters from the muzzle) for the cartridge. This velocity information shall be submitted prior to first article testing. As part of first article testing for conformance to the trajectory, the velocity value will be verified by the Government and established as the velocity requirement for all subsequent production.

3.3.10 Trace. The tracer shall ignite and trace with full luminosity to a range of at least 450 meters. The trace when observed from the firer's position, shall be visible both day and night.

3.3.11 Bullet integrity. The bullet of the cartridge shall not burst or fragment in the barrel or during its effective range when fired.

3.3.12 Ricochet. The bullet or bullet fragments shall not richochet back and endanger the shooter or others located on the firing line, when fired at wood, metal or masonry targets at a distance of 5 meters, and target obliquities between 0 and 30 degrees normal to the line of fire.

3.4 Corrosion. The cartridge shall safely function in the M287 Trainer after being subjected to a 5% salt spray.

3.5 Safety.

3.5.1 Noise. The blast overpressure (noise) produced by the cartridge at the gunner's head position shall not exceed 160 decibles.

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3.5.2 Toxic fumes. The fumes produced by the cartridge shall not exceed 400 parts per million of carbon monoxide.

3.5.3 Maximum range. The maximum range of the cartridge's bullet shall be 1800 meters at standard atmospheric conditions.

3.5.4 Secure cargo (transportation/vibration). After being subjected to the secure cargo (transportation/vibration) test, the test cartridge shall be safe to store, handle and fire.

3.5.5 Sequential rough handling. The cartridges shall be subjected to the following sequential rough handling tests: 2.1 meter (7 foot) drop test; unsecure cargo (loose cargo) test; 1.5 meter (5 foot) drop test; and the 12.2 meter (40 foot) drop test. After the 2.1 meter drop test, the unsecured cargo test and the 1.5 meter drop test, the test cartridges shall be safe to store, handle and fire and the reliability of the cartridges shall not degrade by more than 10%. All cartridges subjected to the 12.2 meter drop test shall not detonate, deflagrate or produce propellant spillage and must be safe to handle and dispose of.

3.5.6 Explosive sensitivity. Explosive sensitivity data shall be generated as specified in 4.5.18 for every explosive material in the cartridge (see 6.5).

3.5.7 Final hazard classification. The cartridge in its shipping and storage package shall meet the following final hazard classification:

DoD Hazard Class:	1.4
DoD Storage Compatability Group:	C
DOT Hazard Class:	Class C Explosive
DOT Container Marking:	Small Arm Ammunition

3.5.8 Material safety sheets. A material safety data statement in accordance with FED-STD-313 shall be prepared for every hazardous material in the cartridge (see 6.4).

3.6 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects. The cartridge shall be free of folds, wrinkles, deep draw scratches, scaley metal, dents and other defects.

3.6.2 Foreign matter. The cartridge shall be free of corrosion, stains, discoloration, dirt, oil and smears of lacquer.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The provisions of MIL-A-48078 shall apply.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective material either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 for definitions of quality assurance terms encountered.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3).

b. Quality Conformance Inspection (see 4.4).

4.3 First article inspection.

4.3.1 First article sample. The contractor shall submit a first article sample of 15,000 cartridges as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall be of the same design as will be submitted for normal production. It shall be manufactured using the same materials and processes as will be

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used in normal production. Equipment utilized to manufacture the first article sample shall embody the process techniques to be used in normal production. The first article sample shall not be custom made in such a way that causes the product quality to be different than that which can be expected in normal production.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II.

4.3.3 Rejection. See MIL-A-48078.

4.3.4 Examination for defects. Examination for critical, major and minor defects shall be performed on a class basis. The sample to be examined shall be drawn in such a way that it is representative of the entire quantity submitted. Non-conforming cartridges shall be rejected and put aside for further examination. Disposition instructions for non-conforming cartridges shall be provided by the contracting officer.

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Table II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-70845(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Cartridge, 9mm, Tracer, M939				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects	800	Acc/Rej	3.1	4.4.4.2
	Critical class		0 - 1		
	Major class		2 - 3		
	Minor class		12 - 13		
	Primer sensitivity	2/	1/	3.3.1	4.5.1
	Residual stress	50	1/	3.3.2	4.5.2
	Bullet extraction	50	1/	3.3.3	4.5.3
	Water proof	50	13 - 14	3.3.4	4.5.4
	Pressure and velocity			3.3.5 & 3.3.9	4.5.5
	+21°C	120	1/		
	-18°C	120	1/		
	+60°C	120	1/		
	Dispersion	300	1/	3.3.6	4.5.6
	Function and casualty			3.3.7	4.5.7
	+21°C	300	Table II		
	-18°C	300	Table II		
	+60°C	300	Table II		
	Trace	100	20 - 21	3.3.10	4.5.9

NOTES:

1/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the first article.

2/ Failure of the rounds to safely function (in accordance with Table III) after the corrosion test shall be cause for rejection.

3/ See 4.4.4.2, Note 2.

AMSMC Form 1570a, 1 Apr 85 Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used.

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Table II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Cartridge, 9mm, Tracer, M939				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			Acc/Rej		
	Trajectory	100	1/	3.3.8	4.5.8
	Bullet intergrity	(150) 3/	1/	3.3.11	4.5.10
	Ricochet	90	1/	3.3.12	4.5.11
	Corrosion	50	2/	3.4	4.5.12
	Noise	10	1/	3.5.1	4.5.13
	Toxic Fumes	150	1/	3.5.2	4.5.14
	Maximum range	4/	1/	3.5.3	4.5.15
	Secure cargo	4000	1/	3.5.4	4.5.16
	Sequential rough handling	5/	1/	3.5.5	4.5.17
	Explosive sensitivity	-		3.5.6	4.5.18
	Final hazard classification	8000		3.5.7	4.5.19

NOTES:

1/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the first article.

2/ Failure of the rounds to safely function (in accordance with Table III) after the corrosion test shall be cause for rejection.

3/ Perform concurrently with function and casualty testing.

4/ Perform concurrently with trajectory testing.

5/ Utilize samples from secure cargo testing.

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TABLE III. Firing defects

	Defects	Acceptance Percentages
a.	Misfire	0.00
b.	Bullet remaining in bore 1/	0.00
c.	Perforation in firing pin indent in primer cup	1.00
d.	Escape of gas around primer cup	
	(1) more than 50% of periphery	0.50
	(2) less than 50% of periphery	1.00
e.	Blown primer 2/	0.00
f.	Loose primer which falls out of pocket	0.00
g.	Loose primer which does not fall out of pocket	0.50
h.	Case split (longitudinal)	
	(1) Mouth (defined as the length of bullet intrusion)	2.00
	(2) Body	0.20
	(3) Through the head	0.00
i.	Case rupture (circumferential)	
	(1) Partial rupture	0.20
	(2) Complete rupture	0.00
j.	Stoppage applicable solely to ammunition	0.00
k.	Bullet stripping	0.00
l.	Other defects (each kind)	0.20
m.	Total of all defects	3.50

1/ No retest permitted.

2/ Primer not in pocket after firing and both head of cartridge case and pocket enlarged and distorted.

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4.4 Quality conformance inspection.

4.4.1 Lot formation. In accordance with MIL-A-48078.

4.4.1.1 Component parts. Unless otherwise specified, component parts shall be homogeneous and of a lot size convenient to the contractor and inspected, tested and accepted by the contractor. The cartridge lot shall contain:

a. Cartridge cases from one interfix.

b. Bullet from one interfix.

c. Primer from one interfix and not more than 2 lots.

d. Propellant from one interfix and not more than 2 lots.

4.4.1.2 Lot identification. Each lot of ammunition shall be identified as to type, caliber and model, as well as a lot number in accordance with MIL-STD-1168.

4.4.2 Rejection. Failure of the cartridges to conform with the applicable requirements shall be cause for rejection of the lot.

4.4.3 Examination. Examination for defects shall be performed in accordance with 4.4.3.1.

a. Major and minor defects. Attributes sampling inspection (ASI) for major and minor defects shall be performed on a class basis in accordance with MIL-STD-105, Inspection Level II. All non-conforming cartridges shall be rejected. See 6.7 for guidance concerning use of sampling tables.

b. Critical defects. Unless otherwise specified, one hundred percent examination shall be performed for all critical defects. If a visual critical defect is found in a sample either just prior to a firing test or after a firing test (and the defect is not due to the firing), the lot shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.3.1	Cartridge, 9mm, Tracer, M939				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
					1/
Critical					
1.	Split case in K, L or M location (see Dwg. 7643674)		100%	3.1	Visual
2.	Perforated Case		100%	3.1	Visual
3.	No primer		100%	3.1	Visual
4.	Cocked primer		100%	3.1	Visual
5.	Inverted primer		100%	3.1	Visual
Major			2/		
101.	Cartridge, corroded or stained, if etched			3.1	Visual
102.	Case, round head			3.1	Visual
103.	Split case in I or J location (see Dwg. 7643674)			3.1	Visual
104.	Case, beveled underside of head			3.1	Visual
105.	No radius on head (rim)			3.1	Visual
106.	Split bullet jacket			3.1	Visual
107.	Loose bullet			3.1	Visual
108.	Loose primer			3.1	Visual
109.	Total length of cartridge, incorrect			3.1	Gage

NOTES:
1/ Refer to MIL-STD-636 for visual defects. In the event of conflict between paragraph of this specification and MIL-STD-636 as to defect classification, the classification specified herein shall apply.
2/ See 6.7 .

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.3.1	Cartridge, 9mm, Tracer, M939				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
					1/
110.	Cartridge profile failure (requiring more than 20 pounds dead weight to insert in profile and alignment gage)			3.1	Gage
111.	Diameter of extractor groove, incorrect			3.1	Gage
112.	Diameter of head, incorrect			3.1	Gage
113.	Thickness of head, incorrect			3.1	Gage
114.	Depth of primer, incorrect			3.1	Gage
115.	Case length, incorrect			3.1	Gage
Minor			2/		
201.	Cartridge, discolored, dirty, oily smeared			3.1	Visual
202.	Case dent			3.1	Visual
203.	Case draw scratch			3.1	Visual
204.	Case scratch			3.1	Visual
205.	Case scaly metal			3.1	Visual
206.	Case wrinkle			3.1	Visual
207.	Case buckle			3.1	Visual
208.	Case bulge			3.1	Visual

NOTES: 1/ Refer to MIL-STD-636 for visual defects. In the event of conflict between paragraph of this specification and MIL-STD-636 as to defect classification, the classification specified herein shall apply.
2/ See 6.7 .

AMSMC Form 1570a, 1 Apr 85 Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.3.1	Cartridge, 9mm, Tracer, M939				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
					1/
209.	Illegible or missing head stamp			3.1	Visual
210.	Defective head			3.1	Visual
211.	Defective mouth			3.1	Visual
212.	Bullet dent			3.1	Visual
213.	Bullet scratch			3.1	Visual
214.	Scaly metal (bullet)			3.1	Visual
215.	Upset (crooked) point (bullet)			3.1	Visual
216.	Flat point (bullet)			3.1	Visual
217.	Nicked or dented primer			3.1	Visual
218.	No waterproofing material (primer pocket joint)			3.1	Visual
219.	Workmanship			3.6	Visual

NOTES:
1/ Refer to MIL-STD-636 for visual defects. In the event of conflict between paragraph of this specification and MIL-STD-636 as to defect classification, the classification specified herein shall apply.
2/ See 6.7.

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4.4.4 Tests. Testing as specified in 4.4.4.2 shall be performed in accordance with the methods specified in 4.5.

4.4.4.1 Test samples. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be throughly mixed before being divided into samples for the various tests.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.4.2	Cartridge, 9mm Tracer, M939				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Primer sensitivity	2/	1/	3.3.1	4.5.1
	Residual stress	50	1/	3.3.2	4.5.2
	Bullet extraction	20	1 - 2	3.3.3	4.5.3
	Waterproof	20	5 - 6	3.3.4	4.5.4
	Pressure and velocity			3.3.5 &	4.5.5
	+21°C	40	1/	3.3.9	
	-18°C	40	1/		
	+60°C	40	1/		
	Dispersion	40	1/	3.3.6	4.5.6
	Function and casualty			3.3.7	4.5.7
	+21°C	150	Table III		
	-18°C	150	Table III		
	+60°C	150	Table III		
	Trace	50	10 - 11	3.3.10	4.5.9

NOTES:

1/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot, subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.4.2	Cartridge, 9mm Tracer, M939				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD

NOTES:
2/ This test shall be performed prior to all other tests cited in 4.4.4.2. Test sample shall consist of the number of cartridges required to perform a complete run-down test wherein 50 primers are tested at each height. If the average critical hight (H) plus three standard deviations (3 sigma) exceeds the all fire requirement or if the average critical height minus two standard deviations (2 sigma) is less than the no fire requirement, the lot shall be rejected, subject to performance of a second complete run-down test wherein 100 primers are tested at each height. Failure of the second sample to comply with the above criteria shall be cause for rejection of the lot.

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4.4.5 Packaging, packing and marking. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644. During or immediately prior to packaging operation, 100 percent examination of the cartridge shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of any incorrect type cartridges shall be classed as a major defect. All non-conforming cartridges shall be rejected.

4.4.6 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See LI12913886, section 6 of MIL-A-48078 and 6.2 herein.

4.5 Methods of inspection. All firings in the trainer system shall be performed utilizing a Government approved firing fixture. Type AA, 40 pound Kraft, Type I wrapping paper (minimum, 1 meter x 1 meter) shall be utilized where witness screens are specified.

4.5.1 Primer sensitivity. The test shall be conducted in accordance with DOD-STD-1468, section 9.

4.5.2 Residual stress.

4.5.2.1 Cartridge with brass cases. The test shall be conducted in accordance with DOD-STD-1468, section 7.

4.5.2.2 Cartridges with polyethylene cases. Test shall be conducted in accordance with Appendix A.

4.5.2.3 Cartridges with steel cases. Test not required.

4.5.2.4 Cartridges with cases of other materials. The test shall be conducted in accordance with procedures approved by the Government appropriate to the material(s) used to manufacture the cartridge.

4.5.3 Bullet extraction. The cartridge shall be tested by the Government approved bullet extraction machine in accordance with DOD-STD-1468, section 5.

4.5.4 Waterproof. The test shall be conducted in accordance with MIL-STD-1168, section 8, procedure II.

4.5.5 Chamber pressure and velocity. The test sample shall be conditioned at $20^{o}C \pm 2^{o}C$, $-18^{o}C \pm 2^{o}C$ and $60^{o}C \pm 2^{o}C$ for at least two (2) hours. The test shall be conducted in accordance with MIL-STD-1168, section 4, except action time need not be measured.

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4.5.6 Dispersion. The test shall be conducted in accordance with Appendix B.

4.5.7 Function and casualty. The test shall be conducted in accordance with MIL-STD-1168, section 6 with the following amendments:

a. The total quantity of round fired shall be as specified in Table II and in 4.4.4.2.

b. Two M287 Trainer Systems shall be used, with one half the total rounds to be fired in each Trainer System.

c. Test cartridges shall be chambered one by one and fired only in single shots in uniform interval.

d. The Trainer Systems shall be fired only in horizontal position.

e. The barrels shall be air or water cooled after every 100 round firing.

4.5.8 Trajectory. Contractor shall submit 100 rounds of M939 practice cartridges to the Government for trajectory testing. (See 6.8.).

4.5.9 Trace. The test shall be conducted in accordance with Appendix C.

4.5.10 Bullet integrity. The test shall be conducted during function and casualty testing (see 4.5.7). Fifty (50) rounds shall be fired at each temperature (20°C, -18°C and 60°C). A paper sheet (6' x 6' minimum) shall be stretched tightly over a rigid frame of the same size. The sheet shall be placed perpendicular to the barrel of the weapon, 15 feet from the muzzle. The sheet shall be replaced as necessary to facilitate observation. Any evidence of bullet fragmentation indicated by irregular perforations or by the number of perforations exceeding the number of rounds fired shall be noted. All irregular perforations shall be measured. All irregular perforations greater than 1/10" shall be classed as defects.

4.5.11 Ricochet. The test shall be performed using the M287 Trainer, fired from a rigid test mount. Ricochet against the following target materials shall be evaluated:

a. Plywood - 1/2 inch thick.

b. Concrete block wall.

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c. Steel plate - 1/4 inch thick.

Each target shall be a minimum of 1 meter wide x 1 meter high. The point of impact of the target shall be 5 meters from the muzzle of the weapon. The target shall be capable of being maneuvered to obtain target obliquities of 0° to 30° (0° representing the condition where the target is perpendicular to the line of fire). A Kraft paper witness screen shall be placed at the muzzle of the weapon, perpendicular to the line of fire. The purpose of the witness screen is to determine if bullets or fragments of bullets will ricochet back into an area where gunners might be located in a typical training scenario. The witness screen shall be a minimum of 10 meters wide x 2-1/2 meters high and shall be located on ground level. The screen will be situated so that there will be 8 meters of paper to the side of the weapon that will indicate ricochets when target obliquity is greater than zero. All firings shall be performed remotely in a single shot mode. The cartridges shall be fired at each target type at 0°, 15° and 30° target obliquities. Equal quantities of test cartridges shall be fired at each target material and at each obliquity angle from the weapon. The screen shall be inspected for perforations greater than 1/10 inches caused by ricocheted bullet material or fragments of the target material shall be cause for rejection. The test shall not be conducted when the wind velocity is greater than 5 miles per hour in any direction.

4.5.12 Corrosion. The sample shall be subjected to the salt-fog test outlined in method 509.2 of MIL-STD-810. After 48 hours of exposure to the 5% mixture the sample shall be examined and its condition documented. The round shall then be chambered and fired in the M287 Trainer. Any defects specified in Table III for which the acceptance percentage criteria is 0.00 shall be recorded.

4.5.13 Noise. The test shall be conducted in accordance with Appendix D.

4.5.14 Toxic fumes. Testing will be performed by the Government in accordance with TOP 2-2-614.

4.5.15 Maximum range. Testing shall be performed by the Government concurrently with trajectory testing.

4.5.16 Secure cargo (transporation/vibration). Testing will be performed by the Government in accordance with TOP 1-2-601.

4.5.17 Sequential rough handling. Testing will be performed by the Government in accordance with TOP 4-2-602.

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4.5.18 Explosive sensitivity. The ARDEC Safety Office shall determine which materials utilized in the cartridge require sensitivity evaluation testing by the contractor. The sensitivity tests required shall be as follows:

Test	Test Document	Test Procedure/ Paragraph
a. Friction Test - (apparatus and comparison values)	MIL-STD-1751	5.4.8 or 5.5.3
b. Impact Test - (apparatus and comparison values)	MIL-STD-1751	5.2.2, 5.4.2 or 5.5.1
	TB 700-2	5.2i
c. Electrostatic sensitivity - (apparatus and comparison values)	MIL-STD-1751	5.2.3, 5.4.7 or 5.5.4
d. Auto ignition temperature	MIL-STD-1751	5.3.9
e. Explosive temperature (5 sec)	MIL-STD-650	506.1
f. Detonation Test -	TB 700-2	5.2f
g. Ignition and unconfined burning test	TB 700-2	5.2g
h. Thermal stability test	TP 700-2	5.2h
i. Card gap test	TB 700-2	5.2j
j. Explosibility of dusts	BOM Rpt S624 pg12	Minimum Explosive Concetration

4.5.19 Final hazard classification. Final hazard classification tests shall be performed in accordance with TB 700-2, paragraph 5.3. The Government reserves the right to observe all final hazard classification tests (see 6.6). The following shall be recorded:

a. Nomenclature and top drawing of the item.

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b. Packaging description of the item and packaging drawings.

c. Test data in accordance with TB 700-2.

d. Photographs before and after test.

4.6 Defect penalty. In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional round shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table III.

4.7 Test validity. If for any reason the test site considers that the test conditions have detrementally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

5. PACKAGING

5.1 Unit pack. Fifty (50) M939 cartridges shall be packed in accordance with dwg. 12913962.

5.2 Packaging. Twenty (20) unit packs shall be packaged in an M2A1 metal ammunition container in accordane with dwg. 12913960.

5.3 Packing. Two (2) M2A1 containers shall be packed in accordance with dwg. 12913959.

5.4 Marking. The unit pack, packaging and packing shall be marked in accordance with dwg. 12913962, 12913960 and 12913959, respectively.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The cartridge covered by this specification are intended for use in the M287 Sub-caliber Trainer for the AT4.

6.2 Acquisition requirements. Acquisition documents shall be in accordance with MIL-A-48078 and should specifiy the following:

a. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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b. Responsibility for furnishing ammunition and Government property.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. The address for submission will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing address for submission of data are listed in FED-STD-313 and below:

Commander
U.S. Army Armament, Munitions
and Chemical Command
ATTN: AMSMC-SFP
AMSMC-TMP-P
AMSMC-PC
AMSMC-AR-ESK
Rock Island, IL 61299-6000

6.5 Explosive sensitivity data . Results of explosive sensitivity testing (see 3.5.5) shall be submitted to Commander, U.S. Army ARDEC, ATTN: SMCAR-SF, Picatinny Arsenal, NJ 07806-5000.

6.6 Final hazard classification. The ARDEC Safety Office shall be notified two weeks prior to the start of hazard classification testing (see 4.5.19). The results of hazard classification testing shall be provided to Commander, U.S. Army ARDEC, ATTN: SMCAR-SF, Picatinny Arsenal, NJ 07806-5000.

6.7 Attributes sampling inspection. Unless otherwise specified in the contract, the contractor may inspect for those characteristics designated as "ASI" in the Classification of Characteristics paragraphs by using sampling procedures in accordance with MIL-STD-105.. Sampling for minor defects should utilize plans indexed by an AQL of 0.65 percent or less (tighter). Sampling for major characteristics should utilize plans indexed by an AQL of 0.25 percent or less (tighter). The AQL is simply an index to the sampling tables in the standard and the designation of an AQL does not imply that the contractor has the right to knowingly supply any defective unit of product. All sampling procedures must be specifically detailed and incorporated into the contactor's quality program plan or detailed inspection system.

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6.8 Trajectory samples. Sample cartridges for trajectory testing shall be submitted to Commander, U.S. Army ARDEC, ATTN: SMCAR-CCL-SD, Picatinny Arsenal, New Jersey 07806-5000.

6.9 Hazard notice. The cartridges described herein and some of its components are flammable and explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to prevent fire, explosion, adverse environment, rough handling, corrosive atmosphere and electrically induced incidents. Such measures shall include the employment of an effective safety program that addresses the inherent hazards associated with the cartridge.

6.10 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or US Army Armament Research and Development Command (ARRADCOM) drawings. Technical data orignally prepared by these activities are now under the cognizance of ARDEC.

6.11 Subject term (key-word) lisiting.

Cartridge, 9mm
Military Specification
Practice Cartridge
Small Caliber Ammunition
Tracer Cartridge

6.12 Certification. This specification is certified to be in compliance with current Army Material Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specification.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-AD17)

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APPENDIX A

RESIDUAL STRESS TEST PROCEDURE

10.0 SCOPE

10.1 Scope. The residual stress test is performed to determine if the residual stress in polyethylene cases is great enough to cause splits or cracks in service or during long term storage.

20.0 APPLICABLE DOCUMENTS- This section is not applicable to this appendix.

30.1 Equipment. Equipment used in this test shall be approved by the Government prior to testing (see 6.3).

30.2 Detergent. Detergent, Nonylphenoxy Poly, (Ethyleneoxy) Ethanol is required for this test.

30.3 Samples. Test samples shall consist of complete sets of the quantity of parts specified with the exception of propellant and primer (insert cartridge). Process the components on the production line and assemble to meet all final assembly dimensional requirements.

40.0 TEST PROCEDURES.

40.1 Immersion. Immerse the test samples in the detergent. Seal the detergent coated samples in a polythylene bag to minimize evaporation and place them in an aluminum tray. Place the tray with the sealed bags of test samples in an oven set at 160°F ± 5°F for seven (7) days.

40.2 Soak. After seven (7) days remove the tray from the oven and cool for one hour. Rinse the insert test samples in running water to remove the detergent. Dry with clean rags or paper towels. Coat the surfaces with machinist's dye and wipe with rags or towels, removing all excess dye from the surfaces.

40.3 Examination. Examine the surfaces from cracks, splits and crazing using a 7 power eye loupe or magnifying glass.

50.0 RECORDING THE DATA.

50.1 Record the following:

a. Date of test.
b. Test ammunition lot number and specification number.
c. Cracks - number, size and location (dwg. 7643674).
d. Splits - number, size and location (dwg. 7643674).
e. Crazing - number, size and location (dwg. 7643674).
f. Technician.
g. Foremen.

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APPENDIX B

DISPERSION TEST PROCEDURE

10.0 PURPOSE.

10.1 Purpose The dispersion test is fired under controlled conditions at a target located at a specified distance from the weapon to determine the dispersion of bullets.

20.0 APPLICABLE DOCUMENTS This section is not applicable to this appendix.

30.0 EQUIPMENT.

30.1 Equipment. Equipment listed in the applicable Inspection Equipment List shall be used (see 6.3).

30.2 Target. The target for Accuracy Tests shall not be less than 36" x 36". The target shall be located 100 yards from the weapon. A windup motor operated by remote control may be used for changing to a new target.

30.3 Mount. In order to minimize weapon inaccuracies, the mount on which the test assembly is mounted, shall be of solid construction.

40.0 PREPARATION.

40.1 Wind velocity. The flight of the bullet is affected to a degree by the direction and velocity of the wind over the outdoor range. Therefore, accuracy tests should not be fired when the velocity of the transverse wind is greater than 8 miles per hour, or varying by more than 4 miles per hour.

40.2 Weapon. The M287 Sub-caliber Trainer shall be used for dispersion testing.

40.3 Test mount. The accuracy test weapon is assembled and clamped in position on the mount and boresighted into position. It is of prime importance that the assembly be mounted properly so that the weapon maintains its original position from shot to shot.

40.4 Test cartridges. The test cartridges shall be placed at a point convenient to the technician. It is not necessary to condition the ammunition at a specified temperature prior to firing, but it is necessary that all cartridges be uniform with respect to temperature.

40.5 Weapon/targets. Two trainers shall be used. The number of targets to be fired shall be divided equally among the two weapons.

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50.0 CONDUCTING THE TEST.

50.1 Warmers. A sufficient number of unrecorded cartridges of the type of ammunition under test shall be fired to assure that the test weapon is correctly sighted on the target, but in any event a minimum of 5 cartridges shall be fired to warm and foul the weapon when it is first put into service and after it has been cleaned or cooled. The approximate location of the shots fired is reported to the technician in order that alignment of the weapon may be adjusted, if necessary.

50.2 Targets. After the warming (fouling) cartridges have been fired, the target shall be changed so as to present a fresh surface for the succeeding target. Thereafter, the target shall be changed after each target of 10 cartridges has been fired.

50.3 Firing rate. Consideration is not given to the position of the propellant in the cartridge case, except that the manner of handling the ammunition from cartridge to cartridge is reasonably uniform. Ten cartridges shall then be fired in even sequence at a rhythmic uniform rate, as rapidly as service of the weapon permits.

50.4 Test barrel. The procedure prescribed in 40.3 shall then be repeated until the specified number of targets has been fired in each trainer. At no time should the exposed metal surface of the test barrel become too hot to grasp with the bare hands.

60.0 ANALYSIS OF TARGET.

60.1 Measurement of the target. The vertical reference axis and horizontal reference axis for each target is drawn onto the target. A vertical line drawn through the center of the left-most shot hole represents the vertical reference axis. A horizontal line drawn through the center of the bottom-most shot hole represents the horizontal reference axis. For each target, the vertical distance from the horizontal reference axis to each shot hole is measured, recorded and used in calculating the vertical standard deviation for that target. For each target, the horizontal distances from the vertical reference axis to each shot hole is measured, recorded and used in calculating the horizontal standard deviation for that target. The shot holes should be numbered and the measurements taken in sequence to aviod confusion.

60.2 Calculation. From these measurements, the vertical and horizontal standard deviations from each group of shots are calculated using the following formula:

$$\text{Standard Deviation} = \frac{n \quad d^2 \quad - \quad (\quad d)^2}{n \ (n-1)}$$

Where n = the number of shot holes
d = distance from reference axis

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60.3 Target misses. Should a target not contain 10 shot holes, proper alignment of the weapon shall be verified by calculating the location of the center of impact of the shot holes on the target. The target shall be considered invalid (and must be refired) if the center of impact is too near the edge of the target (13" or less from the nearest edge of the target). Otherwise, the target is valid and the missing shot holes shall be counted as defects against the lot, in accordance with the specification.

70.0 RECORDING OF RESULTS.

70.1 Test results. All test results shall be recorded to the nearest one-tenth of an inch.

70.2 Test sheets. Test sheets shall show the following:

a. Vertical standard deviation.
b. Horizontal standard deviation.
c. Velocity and direction of wind.
d. ALL FUNCTION AND CASUALTY DEFECTS
e. Failures to trace.

80.0 FACTORS AFFECTING ACCURACY TESTS.

80.1 External factors. The external factors should be controlled as closely as possible in order to obtain results that are a representative of the inherent accuracy of the ammunition.

80.2 Test weapon. The dimensions and condition of accuracy weapons, as well as the manner of placing the weapon in the rest, are of prime importance.

80.3 Handling. The technique of test can affect accuracy results. If uniformity in the manner in which the cartridges are handled, chambered and fired is poor, a larger spread is obtained than if the cartridges are tested in a uniform manner.

80.4 Barrel temperature. The temperature of the test barrel should be controlled.

80.5 Atmospheric conditions. The atmospheric conditions existing at the time the test is fired can influence the test results.

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APPENDIX C

TRACE TEST PROCEDURE

10.0 PURPOSE.

10.1 Purpose. To determine the relative effectiveness of tracer ammunition to disclose the path of bullet flight by firing it under controlled conditions, and determine thereby the acceptance or rejection based on the criteria specified in the applicable specification.

20.0 APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30.0 EQUIPMENT.

30.1 Equipment. Equipment used in this test shall be approved by the Government prior to testing (see 6.3).

30.2 Firing range. Firing range shall be of sufficient length to permit firing of cartridges in accordance with this specification.

40.0 OBSERVATION POINTS.

40.1 Observation for night testing. Trace performance shall be made at the weapon and at points 65 meters and 450 meters beyond the muzzle of the gun on a line parallel to and approximately 70 meters from the plane of the trajectory. Lights or other suitable visible markers shall be placed opposite the 65 meters and 450 meters observation points to insure a line of sight from each point perpendicular to the plane of the trajectory.

40.2 Observation for day testing. The observer at this point shall observe for any abnormal behavior as an informational check only. Record shall be made of erratic flight, bursting bullets, muzzle flash of obviously greater intensity than that produced by other bullets in the sample, or such other behavior as may be considered abnormal by the observer.

40.3 Observation at 65 meter position. The observer, with line of sight established perpendicular to the trajectory by the light or the visible marker, shall note the trace performance of the bullet from each sample cartridge. A trace which is visible at 65 meters shall be considered acceptable. The cartridge shall be considered defective if trace is not visible at 65 meters.

40.4 Observation at 450 meter position. The observer, with line of sight established perpendicular to the trajectory by the light or the visible marker, shall note the trace performance of the bullet from each sample cartridge. A trace which is visible at 450 meters shall be considered acceptable. The cartridge shall be considered defective if trace is not visible at 450 meters.

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50.0 TEST PROCEDURES.

50.1 Preparation for firing.

50.1.1 Trace test. Trace test shall be conducted in accordance with the requirements contained herein and in this specification. The trace testing shall be performed at night.

50.1.2 Barrels. Barrels used for trace testing shall be used for trace tests only. The test barrel shall be at ambient temperature prior to firing each test.

50.1.3 Test mount. The Trainer System is assembled in the fixture on the mount. The pier or mount shall be of solid construction.

50.1.4 Firing sequence. Since temperature conditioning of the ammunition is not necessary, the ammunition shall be fired under existing conditions; however, to prevent temperature of weapon from affecting cartridge performance, the cartridges shall be fired in a uniform sequence.

50.2 Firing the test.

50.2.1 Warmers. Three unrecorded cartridges shall be fired to sight and warm the weapon prior to each test.

50.2.2 Firing sequence. The cartridge shall be fired in a uniform sequence, with sufficient interval between cartridges to allow the trace of each individual shot to be observed and recorded by each observer.

60.0 RECORDING THE DATE.

60.1 Trace characteristics. The trace characteristic of each shot fired are recorded as it passes each observation point. After the test is completed, the observer must check their respective observations together, shot by shot. Defects reported at more than one point for the same shot shall be recorded as one failure.

60.2 Record the following:

a. Date of test.
b. Test ammunition lot number and specification number.
c. Number of cartridges fired.
d. Satisfactory trace (percentage).
e. Number and type of defects.
f. Number and type of case casualties.
g. Technician.
h. Foreman.

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60.3 Weapon test data. The following test weapon data shall be recorded on the test sheet.

a. Receiver number.
b. Barrel number.
c. Total number of cartridges fired in barrel prior to test.

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APPENDIX D

NOISE LEVEL TEST PROCEDURE

10.0 PURPOSE.

10.1 Purpose. To measure the noise level of cartridges at the gunner's head position.

20.0 APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30.0 EQUIPMENT.

30.1 Equipment. Equipment used in this test shall be approved by the Government (see 6.3) prior to testing.

30.2 M287 Trainer.

40.0 TEST PROCEDURES.

40.1 Pre-firing (preparation for test).

40.1.1 Temperature conditioning. Test cartridges shall be conditioned at $70^{o} \pm 2^{o}F$ for a minimum of four hours.

40.1.2 Trainer. The trainer shall be assembled in the test fixture on the mount.

40.1.3 Microphone #1. A noise level meter is set up outside the firing room with a microphone placed 1 meter to the right of the trainer.

40.1.4 Microphone #2. A second microphone is placed at the right handed gunner's head position.

40.2 During firing.

40.2.1 Observer. An observer is stationed at the noise level meter.

40.2.2 Firing. The test cartridges shall be loaded into the trainer and fired.

40.2.3 Noise levels. The noise levels are recorded.

40.2.6 Case examinations. Fired cases shall be visually examined by the technician for possible case casualties.

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50.0 RECORDING THE DATA

50.1 Record the following:

a. Date of test.
b. Test ammunition lot number and specification number.
c. Ammunition temperature.
d. Gun room temperature.
e. Firing range temperature.
f. Outdoor temperature.
g. Gun type and serial number.
h. Number of times the barrel has been fired.
i. Gun headspace.
j. Firing pin protrusion.
k. Firing pin indent.
l. Case casualties.
m. Any abnormality.
n. Gunner.
o. Foreman.
p. Location of microphones.

50.2 Computations. Calculate the difference in noise levels between the cartridge and the requirement.

50.3 Results. Show all the information on the data sheets plus the calculated noise level differnce.

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	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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